

STAVROPOL, Russian Federation

WMO: 349490

Lat: 45.1084N

Lon: 42.0999E

Elev: 1481

StdP: 13.93

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	8.3	-3.6	4.8	1.6	3.5	6.9	9.9	32.4	33.6	28.7	30.2	9.7	90	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	92.9	66.9	89.6	66.2	86.2	65.5	70.6	84.6	69.1	82.3	67.7	80.6	12.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	102.1	75.9	64.6	96.2	74.6	62.9	90.7	73.4	35.4	84.9	34.1	82.7	33.0	80.4	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.1	24.2	21.1	-1.0	96.8	9.2	3.2	-7.6	99.2	-13.0	101.0	-18.2	102.9	-24.9	105.2		
			-1.8	73.0	8.9	2.0	-8.1	74.4	-13.3	75.6	-18.3	76.7	-24.7	78.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	28.6	30.2	38.8	50.1	60.3	68.7	73.7	73.6	63.5	51.3	39.8	32.2
	DBStd	18.07	10.27	10.55	8.09	7.62	6.65	5.94	5.15	5.84	7.08	9.00	9.24	9.29
	HDD50	2643	663	555	357	88	6	0	0	3	94	325	552	
	HDD65	5880	1127	974	812	447	173	28	3	107	431	756	1016	
	CDD50	3020	0	1	11	93	325	561	736	730	408	134	19	2
	CDD65	781	0	0	0	1	27	139	274	272	62	6	0	0
	CDH74	7333	0	0	2	18	241	1173	2610	2713	523	53	0	0
	CDH80	2756	0	0	1	2	36	377	1033	1150	143	14	0	0
Wind	WSAvg	10.1	10.2	10.7	11.6	11.0	9.6	9.3	9.2	9.4	9.9	10.1	10.2	10.5
Precipitation	PrecAvg	21.7	1.1	1.1	1.2	1.8	2.5	3.2	2.2	2.0	1.7	1.8	1.6	1.5
	PrecMax	31.4	3.7	2.9	2.7	4.2	5.2	8.6	6.3	6.6	4.3	4.8	3.4	3.9
	PrecMin	14.9	0.1	0.0	0.1	0.6	0.0	0.0	0.1	0.2	0.3	0.0	0.1	0.2
	PrecStd	4.1	0.7	0.7	0.7	0.9	1.2	2.0	1.4	1.5	0.9	1.2	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	58.7	68.3	76.4	84.1	91.7	95.3	96.6	89.9	80.7	68.3	57.6
		MCWB	43.5	47.1	51.0	56.7	64.4	66.4	68.6	66.8	65.2	60.6	51.5	46.0
	2%	DB	46.7	52.1	61.1	71.6	80.2	87.9	91.8	93.3	84.0	73.3	61.1	51.6
		MCWB	40.2	44.0	47.9	55.4	62.5	65.4	67.0	66.9	64.1	57.8	49.2	43.3
	5%	DB	43.1	46.8	55.6	67.6	76.6	84.3	89.5	89.9	79.8	68.2	55.8	46.8
		MCWB	38.6	40.7	45.7	54.0	61.1	64.8	66.7	65.9	61.9	56.8	47.8	41.3
	10%	DB	39.6	42.7	51.4	63.0	73.1	80.7	86.0	86.3	75.6	64.3	51.9	43.1
		MCWB	36.2	38.3	43.8	51.5	60.1	64.7	66.0	65.2	60.6	55.4	46.3	39.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	48.5	52.4	59.2	67.0	71.8	72.9	72.4	68.7	62.1	54.0	47.8
		MCDB	50.5	55.7	64.9	71.5	79.5	83.8	88.3	87.5	81.6	73.5	61.8	55.6
	2%	WB	41.4	44.5	49.1	56.7	64.5	69.2	70.7	70.4	65.9	59.9	51.5	44.5
		MCDB	45.3	51.5	58.5	67.9	76.3	81.2	85.2	84.6	78.0	69.7	59.1	49.2
	5%	WB	38.7	41.3	46.8	54.8	62.7	67.3	69.0	68.9	64.0	58.1	49.0	41.9
		MCDB	42.1	46.3	54.0	64.7	73.0	79.0	82.6	82.2	75.0	66.2	54.9	45.6
	10%	WB	36.6	38.2	44.3	53.0	60.9	65.6	67.6	67.5	62.5	56.1	46.6	39.8
		MCDB	39.3	41.8	49.7	62.0	70.7	77.1	80.9	80.5	72.5	63.0	51.3	43.0
Mean Daily Temperature Range	5% DB	MDBR	11.1	12.5	14.8	17.9	17.7	18.3	19.8	20.8	18.6	15.4	12.7	11.2
		MCDBR	15.6	18.5	23.2	23.4	23.0	23.2	24.3	25.3	24.1	22.0	20.0	16.3
		MCWBR	10.9	11.9	13.0	12.0	11.3	10.0	9.6	9.9	10.0	10.9	11.8	11.0
	5% WB	MCDBR	14.1	17.4	20.5	20.2	20.2	19.7	20.8	21.2	20.3	19.2	17.5	14.6
		MCWBR	10.5	11.7	12.4	11.3	10.7	10.2	9.6	9.6	9.7	10.3	11.2	10.5
Clear-Sky Solar Irradiance	taub	0.324	0.356	0.398	0.442	0.409	0.416	0.440	0.439	0.407	0.373	0.338	0.326	
	taud	2.303	2.225	2.158	2.072	2.254	2.275	2.218	2.218	2.277	2.335	2.371	2.316	
	Ebn at Noon	244	257	261	259	273	271	262	258	256	248	238	230	
	Edh at Noon	26	34	41	49	42	42	43	42	36	30	24	23	
All-Sky Solar Radiation	RadAvg	409	685	966	1469	1826	2016	2028	1827	1340	826	469	333	
	RadStd	72	107	93	147	189	117	107	95	137	107	59	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	+243	N/A

Nomenclature: See separate page